

Introduction

In the past, the automotive industry stood as a testament to the power of combustion engines and the prestige of owning a car with the “most exhaust pipes.” Today, this old-school paradigm is undergoing a seismic transformation. Four major innovations—electrification, automation, shared mobility, and connected mobility—are happening all at once, leading to dramatic changes in the automobile landscape. Further amplifying this complexity are industry newcomers who are not just building top-quality cars but are specifically focusing on the digital-savvy consumer with their “smartphones on wheels”—vehicles equipped with large interactive screens, seamless connectivity, and frequent updates to software features.

At the same time, customers are expressing dissatisfaction because their vehicles lack the features and user-friendliness that are commonplace in smartphones. Many are left wondering: why can’t my \$50,000 car perform the same tasks as my \$300 smartphone?

From this frustration emerged the idea of a software-defined vehicle (SDV), a car that’s fully programmable. New features can be developed and deployed within a matter of months, not years, and there’s extra computational capacity for future updates that can be delivered wirelessly. This is a car that keeps getting better, thanks to the continuous delivery of valuable new software features and updates throughout its lifetime.

However, the automotive industry has hurdles to overcome if it wants to achieve this goal quickly. The complexity of software is spiraling out of control, with the volume of code reaching an

estimated 300 million lines with level 5 autonomous driving, all while developers cling to the software development methodologies of yesteryear. Further, hardware and software are still tightly integrated and released simultaneously, meaning significant changes occur only once every seven years or so.

Consider the mobile phone industry, where former leading brands like Nokia and BlackBerry were swiftly outpaced by smartphones. A similar fate for the automotive industry's incumbents seems possible. At first, many veterans of the cell phone industry insisted that their business was fundamentally different from the computing industry, but the smartphone revolution quickly shattered that illusion.

This shift raises several critical questions: How can the incumbents in the automotive industry succeed in their ongoing digital transformation to avoid a similar fate? What makes change so challenging within this industry, and how can we significantly accelerate automotive innovation? What will the future of revenue generation look like? Will revenue still come predominantly from car sales, or will digital services layered on top become the primary revenue driver? And crucially, how do we pinpoint the “killer apps”—those standout applications or digital features that prove so essential or attractive that they drive the success of future vehicle generations?

While the SDV is a key technical enabler, the path to answering these questions starts with adopting a “digital first” strategy. This means starting with the digital customer experience and working backward to the solution designs, engaging in early exploration, and testing to ensure that the vehicles on the road align with the ever-evolving needs of the modern consumer.

Goals of the SDV

Let us start our discussion by looking at why we want to build an SDV in the first place. It isn't about the bits and bytes or the number of sensors and cameras—it's about the customer experiences we build. And these customer experiences cannot aim simply to rebuild the smartphone experience. This is about creating a “habitat on wheels” powered by cross-domain applications and data fusion and delivered and constantly improved at 10 times the speed we see today.

Smartphone Habitat on Wheels

During the past decade, many car manufacturers have sought to replicate successful smartphone applications in their cars. In many cases, however, these in-vehicle applications could not match the quality of the smartphone apps. In addition, consumers usually don't want redundant experiences, inconsistencies between their digital ecosystems, or the irritation of cumbersome data synchronization. Today, car makers have to walk a fine line between application and data ownership on the one hand and customer experience and tighter integration with the dominant smartphone ecosystems on the other.

This is why it is important that the SDV surpass the concept of a “smartphone on wheels.” Instead, it has to enable a “habitat on wheels,” utilizing the specifics of the car to provide multisensory experiences that a smartphone could never match. With multiple displays and a network of hundreds of sensors and actuators, the SDV brings together domains like infotainment, autonomous driving, intelligent body, cabin and comfort, energy, and connected car services, crafting a unique journey.

Passengers feel recognized as they enter a vehicle personalized to their needs, one that is a clear departure from the impersonal confines of traditional cars. Passengers experience a reassuring sense of security while journeying under the vigilant care of cutting-edge safety systems. Passengers step into the vehicle with a green conscience. Passengers reclaim their time, transforming travel into an opportunity for meaningful engagement, whether through work, rest, or play. Passengers with disabilities can enjoy a newfound freedom and mobility thanks to autonomous driving. This technology allows them to move with confidence, their independence undiminished. It's not just about the journey; it's about the freedom to explore, to engage, and to live life unrestricted.

SDVs have revolutionized our perception of mobility. It's no longer merely about getting from point A to point B but about making the journey itself enriching. Thanks to advanced driver-assistance systems and autonomous driving, we're embracing the transformative, multisensory power of the SDV.

In the SDV, the digital and physical worlds merge, creating experiences that are both immersive and intuitive. This fusion allows the car to interpret and react to physical cues with precision and sophistication, enhancing the overall experience. For instance, the vehicle's sensors could identify the state of health of children in the back seat and estimate potential motion sickness based on road curvature and speed. Responding to these subtle cues, the system could adapt the automated driving style, adjust window positions, and regulate the air conditioning to enhance comfort and minimize motion sickness.

Moreover, value-added digital features transform the vehicle into a hub of productivity and entertainment. Imagine video conferences displayed on the front car window as holograms, engaging in interactive games with other drivers during traffic jams, or learning a new language based on your surroundings during your commute—all within the safety and comfort of your personal, mobile space. This seamless blend of the digital and physical worlds, facilitated by the SDV, ensures that every journey is not just a ride but an enriching experience that adds value far beyond arrival at your destination.

Cross-Domain Applications and Data Fusion

Today, vehicle experiences often occur in isolated domains. But the future with SDVs promises to blur the boundary between the vehicle and the outside world. Experiences will be cross-domain, where various vehicle functions and systems intercommunicate and interact harmoniously to enrich the overall journey. As [Figure 1-1](#) shows, SDVs represent a shift from disparate functions to integrated experiences, where the vehicle works as a unified entity rather than a collection of separate parts.

Consider the example of a digital “dog mode” that some cars already feature. The vehicle monitors your dog in the car while you are out shopping. Because it is hard for dogs to cool down, a hot car interior on a summer's day is often enough to cause serious injuries or even death. This is a perfect illustration of customer-centric and cross-domain functionality. It involves multiple systems: the car's air conditioning to maintain a comfortable temperature, the infotainment screen to display a message letting passers-by know not to worry as the dog is safe and comfy, and the battery management system to ensure the car has sufficient energy. All these domains are coordinated in order to ensure the dog's safety and comfort.

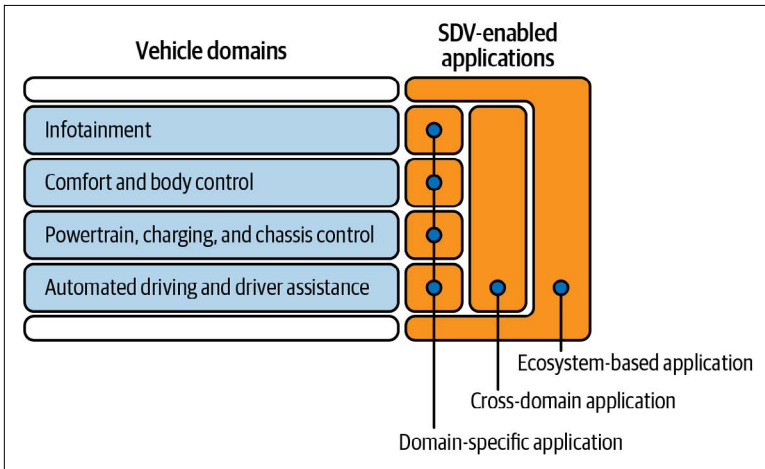


Figure 1-1. Cross-domain application services

In this connected ecosystem, your car could even become a creative extension of your social media presence. With your permission, it could capture a stunning sunset through its high-quality on-board cameras during a scenic drive and propose a pre-edited post for your approval. Your vehicle becomes an active participant in your digital life, enabling you to share unique moments without detracting from the driving experience.

Envision your electric vehicle communicating with a smart grid, scheduling its charging during off-peak electricity usage. This interaction optimizes energy use and maximizes cost efficiency. Your car is no longer just a consumer of energy but also an intelligent participant in the wider energy ecosystem.

Cross-domain experiences also extend to personal wellness. Imagine that your fitness wearable signals that you've had an intense workout. In response, your car sets the cabin temperature to a cooler setting, selects soothing illumination for the ambient lighting, and plays your favorite cool-down playlist. By seamlessly integrating with your digital devices, your car enhances your post-workout recovery and comfort.

This fusion of digital and physical experiences makes the car an extension of your digital life, connecting you seamlessly to the world around you.

Accelerate Innovation by 10x

To achieve the customer-centric development and continuous improvement required to support the usability goals outlined above, development of SDVs must significantly speed up, shortening the time to market for new features. **Figure 1-2** shows the transformation that must occur.

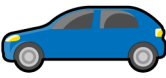
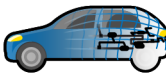
	 Today's typical vehicle	 Software-defined vehicle
Development cost	\$\$\$\$	\$
First prototype	Months	Hours
Time to market	3 years	Weeks to months
Approach	First-time right at HW SOP	MVP and continuous improvement
Scalability	Every car model needs different code	Same code for all car models

Figure 1-2. “Need for speed” in automotive software development

The goal here should be a tenfold improvement over the old ways of delivering new features. Development costs must be significantly reduced. Initial prototypes must be available in hours, not months. Time to market must be reduced from years to a few weeks.

And all of this does not only apply to new features: we must be able to constantly monitor how customers are using existing features, learn how to improve them, and then optimize them—again, all done 10 times faster than we do it today.

Impediments: Why Is Automotive Software Development Different?

To achieve our goals for the SDV, we have to understand the impediments we need to overcome. Why is automotive software development different from development in other domains (e.g., smartphones)? Sure, “smartphone on wheels” is an accessible metaphor that illustrates the evolving nature of SDVs. With the rise

of touchscreens, robust connectivity, and a host of app-driven functions in our cars, drawing a parallel with our handheld smart devices seems apt. However, the simplistic charm of this metaphor can be misleading, as it does not account for the intricacies and unique challenges of the automotive domain.

The following discussion provides a deep dive into the key impediments to the broad adoption of SDV in the automotive industry, including complexity and heterogeneity, functional safety, the “clash of two worlds,” and the need for organizational transformation.

Complexity and Heterogeneity

A key difference between the automotive industry and the smartphone industry is their levels of complexity. Over the past decade, the smartphone industry has managed to address technical complexity through standardization and abstraction. Today’s smartphones are highly integrated with multiple layers of abstractions and interfaces. The automotive industry, on the other hand, is still plagued by high levels of complexity and heterogeneity.

Furthermore, high levels of abstractions have enabled the smartphone industry to encapsulate most of the hardware complexity, allowing it to deal with complexity on the software level. On the other hand, today’s vehicles are still complex systems of systems. Each subsystem in a car, from the brakes to the powertrain, is a complex entity, supplied by a different vendor and integrated with a unique software architecture. The level of complexity and the need for seamless interoperability among systems far surpasses what we see in today’s smartphones.

Another key difference is that the automotive industry is dealing with many more product variants. Many cars are configured according to individual customer preferences expressed during the sales process. Regional differences in customer preferences and regulatory requirements lead to large numbers of product variants. And different functional requirements—think family van versus convertible versus truck—lead to many different vehicle types, models, editions, and variants. Thus, there are usually much lower production numbers per vehicle model than per smartphone model.

Functional Safety

The “smartphone on wheels” comparison also falls short when we consider automotive functional safety requirements. Unlike smartphones, vehicles are subject to stringent safety standards, such as ISO 26262. This standard deals with the functional safety of electrical and electronic systems within vehicles and is fundamental to the concept of an SDV.

The ISO 26262 standard doesn’t just offer a set of rules but embodies a risk-based philosophy that assesses potential hazards and requires appropriate design techniques to mitigate them. Risks are categorized according to three factors: severity (the potential harm that can occur), exposure (the probable frequency of the risk situation), and controllability (the ability of the driver to prevent the hazard). By evaluating these factors, the standard effectively balances innovation and safety in automotive design.

To quantify the risk, the standard employs a framework known as Automotive Safety Integrity Levels (ASIL), illustrated in [Figure 1-3](#), which classifies hazardous events that could result from a malfunction based on their level of severity, exposure, and controllability. Levels of risk range from ASIL A, the lowest level, to ASIL D, the highest level. ISO 26262 defines the requirements and safety measures to be applied at each ASIL.

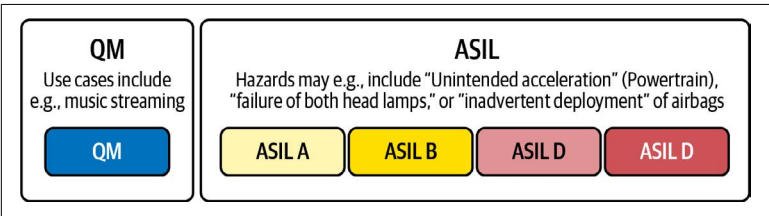


Figure 1-3. Automotive safety levels

Additionally, a class called QM (Quality Managed) is assigned to systems that, after thorough analysis, require only standard quality management processes due to their lower potential impact on safety. For instance, a safety-relevant function like a braking system would be assigned a high ASIL rating, and an in-car entertainment system would be assigned a QM rating. The software controlling the brakes would need adhere to more rigorous safety standards than the entertainment system. The overarching objective in vehicle design is risk mitigation, a process that seeks to reduce potential hazards.

For example, the risk of a false speed signal due to a defective sensor can be mitigated by introducing a redundant sensor. This additional sensor can cross-validate signals, thus minimizing the chances of error and enhancing the overall safety of the vehicle.

Therefore, while the “smartphone on wheels” analogy succinctly portrays the emerging role of software in vehicles, it does not wholly capture the stringent safety standards and rigorous risk mitigation strategies employed in the automotive industry. The SDV is more than a simple mobile device; it’s a sophisticated ensemble of systems that prioritizes safety as much as functionality and convenience.

Clash of Two Worlds

Automotive engineering has traditionally focused on physical functionality, from early electronic features (such as airbags, vehicle stabilization, and braking systems) to modern driver assistance systems or even automated driving. The new, digital experience is driven by software-enabled features, and the focus is on improving the experience of using the vehicle or even providing value-added digital features like usage-based insurance or automated payment of parking fees. [Figure 1-4](#) shows some of these differences.

The traditional world of engineering physical vehicle features is characterized by a complex system of systems—conforming to functional safety, security, and real-time requirements—and homologation, a process of obtaining government-approved certification for product market readiness to ensure that safety and environmental standards are met. In contrast, the new digital vehicle experience is built on best practices from the technology sector, including Agile methods and cloud-style development.

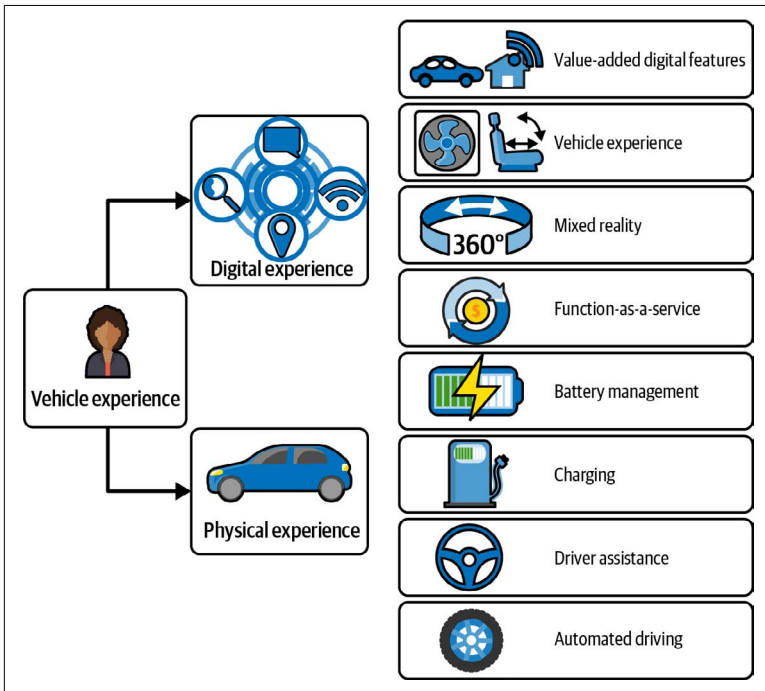


Figure 1-4. Digital versus physical experience

Modern vehicles are a fusion of these worlds; they must merge the reliability of traditional engineering with the agility of modern software development (see [Figure 1-5](#)). This presents the automotive industry with a significant challenge: OEMs that are best able to address this “clash of two worlds” are most likely to succeed.

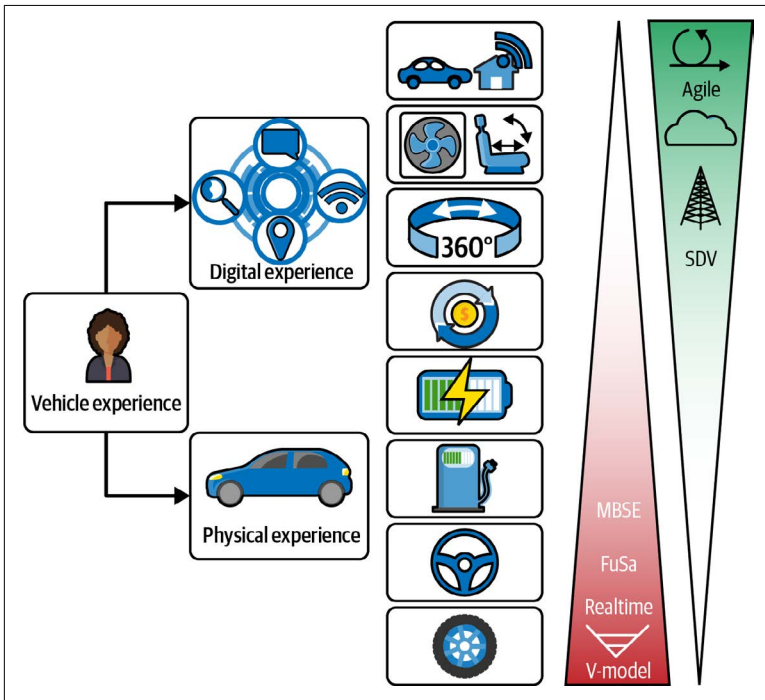


Figure 1-5. Clash of two worlds

Organizational Transformation

We can observe that over the last decade, the incumbent OEMs have undergone complex and large-scale organizational transformation to deliver both physical and digital car features. It is vital that OEMs enable their organizations to combine the worlds of traditional engineering and modern software development and have them work together in relative harmony. But this means that they must create organizations with suborganizations that can move at different speeds and work with different cultures, methods, and tools. It is not trivial to create vehicle platforms and architectures that are so modular that components with different requirements (e.g., functional safety) can be assigned to the organizational units that are the best fit.

This doesn't stop with the company's internal organization. Traditionally, many OEMs purchased different subsystems from different vendors, usually combining hardware and software for each subsystem. With the SDVs, however, the decoupling of hardware and

software must be considered with respect to sourcing. Established supply chains are undergoing significant transformations. The roles of existing vendors are changing, and new vendors are entering the playing field.

Rethinking the Vehicle Lifecycle: Digital First

What can we learn from the discussion of impediments that can be applied to the vehicle lifecycle? Historically, the lifecycle of a vehicle was defined by the simultaneous production and deployment of tightly coupled hardware and software. Once the vehicle was in the consumer's hands, its features remained essentially unaltered until its end of life. However, an SDV paradigm allows for the decoupling of hardware and software release dates—a prerequisite for a digital first approach, which puts design and virtual validation of the digital vehicle experience at the start of the lifecycle.

Digital first means that new ideas for the vehicle experience are initially explored in virtual environments, getting early user feedback long before any custom hardware must be developed or even a physical test vehicle is provided. Digital first means that the principles of design thinking and Lean startup, which originated in internet culture, can now be applied in the physical world of automotive development.

In the SDV model, new ideas are generated with approaches like design thinking, where features are explored virtually and tested with early customer feedback, facilitating a faster and more cost-effective way to find a product–market fit. *Product–market fit* refers to a state in which a product meets the specific needs and desires of its target market—in this case for new vehicle applications and services. The ability to rapidly explore, innovate, and adapt is a cornerstone of the new automotive era.

A lot of this relates to our smartphone discussion. Just as the launch of the iPhone opened doors to applications beyond our wildest dreams—identifying plant species, optimizing sleep quality, studying for a driver's license—the advent of the SDV propels us into a future of unimagined possibilities. The applications and experiences made possible by these intelligent habitats on wheels remain largely untapped, inviting creative minds to explore them.

We can't precisely predict what these revolutionary applications might be. It's likely that the SDV's "killer app" won't just be an equivalent of *Angry Birds* for the car but something that leverages the unique potential of this new mobile environment.

But we can confidently make the following statement: to discover these groundbreaking applications and experiences, experimentation and innovation at scale need to happen. This requires much faster development, as we discussed earlier. However, it also requires a new approach to validating new ideas, especially from the point of view of desirability and usability. Building physical test vehicles is very expensive and time-consuming. Because of this, the automotive industry is looking at ways to virtualize usability and acceptance testing. For example, the digital.auto playground provides an environment that allows us to try out new ideas for digital vehicle features in a pure cloud environment and evaluate them against real vehicle APIs. The emergence of spatial computing and virtual reality will accelerate virtualized UX testing. It is important that these tests not be limited to the physical vehicle design. The ability to test the vehicle experience as it is enabled by SDV in virtual environments will help significantly left-shift user testing. This in turn will help make sure that investments are prioritized according to market demand, and the resulting vehicle experience creates popular products and high levels of customer loyalty. Figure 1-6 summarizes the digital first vehicle lifecycle.

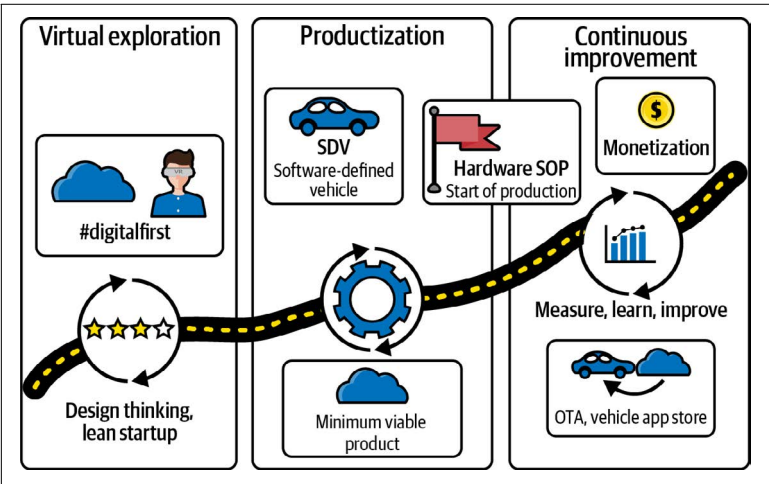


Figure 1-6. Digital first vehicle lifecycle

Unlike in the past when car manufacturers had limited insight into the actual use of certain features, SDVs can, provided the driver gives consent, measure and assess actual usage of vehicle apps and their features. This data-driven approach fosters a fast learning cycle, facilitating innovations that are tailored to customer use and preferences. Chapter 5, “#digitalfirst: A New Way of Working” provides more details on how all of this can be applied in practice.